

ISLP Chapter 12 Exercises: Unsupervised Learning

Questions 8 and 13

MUSABYIMANA Diane
101218
Assignment

Question 8

In Section 12.2.3, a formula for calculating PVE was given in Equation 12.10. We also saw that the PVE can be obtained using the `explained_variance_ratio_` attribute of a fitted `PCA()` estimator. On the `USArrests` data, calculate PVE in two ways and confirm they agree.

(a) PVE via `explained_variance_ratio_`

The `USArrests` data (50 states, 4 variables: Murder, Assault, UrbanPop, Rape) was standardized to mean zero and standard deviation one, then PCA was fitted. The resulting proportion of variance explained by each component was read directly from `explained_variance_ratio_`.

(b) PVE via Equation 12.10 directly

Equation 12.10 defines the PVE of the m -th principal component as:

$$\text{PVE}_m = \frac{\sum_{i=1}^n \left(\sum_{j=1}^p \phi_{jm} x_{ij} \right)^2}{\sum_{j=1}^p \sum_{i=1}^n x_{ij}^2}$$

The loadings ϕ_{jm} were taken from the `components_` attribute of the fitted `PCA()` estimator, and applied to the same standardized data used in part (a).

Component	PVE (method a)	PVE (method b)
PC1	0.6201	0.6201
PC2	0.2474	0.2474
PC3	0.0891	0.0891
PC4	0.0434	0.0434

Result: The two methods give identical results (confirmed with `np.allclose`, which returned `True`). This is expected, since `explained_variance_ratio_` is computed internally using essentially the same calculation as Equation 12.10 — the proportion of total variance captured by projecting the data onto each principal component direction. The first principal component alone explains about 62% of the total variance, and the first two together explain about 87%.

Question 13

A gene expression data set (`Ch12Ex13.csv`) consists of 40 tissue samples with measurements on 1,000 genes. The first 20 samples are from healthy patients, and the second 20 are from a diseased group.

(a) Load the data

The data was loaded with `pd.read_csv(..., header=None)`, giving a 1000×40 matrix: 1,000 genes (rows) by 40 samples (columns), with the first 20 columns labeled H1--H20 (healthy) and the last 20 labeled D1--D20 (diseased).

(b) Hierarchical clustering with correlation-based distance

Correlation-based distance ($1-r$, where r is the Pearson correlation between two samples' expression profiles across all 1,000 genes) was computed between the 40 samples, and hierarchical clustering was applied using three linkage methods: complete, average, and single.

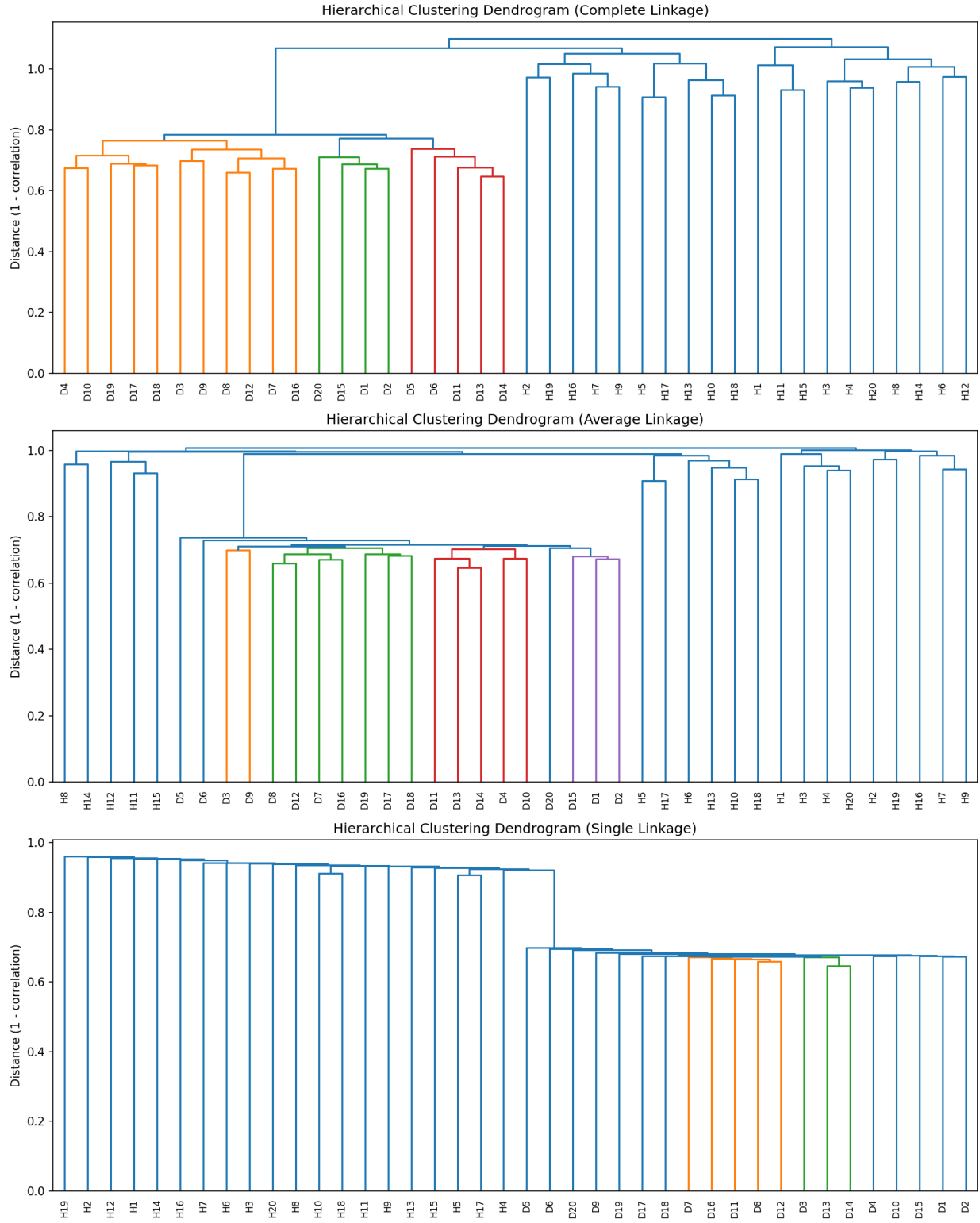


Figure 1: Hierarchical clustering dendrograms for the 40 tissue samples, using correlation-based distance with complete, average, and single linkage.

Cutting each dendrogram at $k = 2$ clusters and cross-tabulating against the true healthy/diseased labels gives:

Linkage	Cluster 1 (D / H)	Cluster 2 (D / H)
Complete	20 / 10	0 / 10
Average	20 / 11	0 / 9
Single	20 / 19	0 / 1

Do the genes separate the samples into the two groups? Only partially, and the outcome clearly depends on the linkage method used. Complete and average linkage both identify a distinct sub-cluster made up entirely of healthy samples (10 and 9 samples respectively), while all 20 diseased samples end up grouped together with the remaining healthy samples in the other cluster — so the split is not a clean 20-vs-20 separation matching the true labels. Single linkage performs far worse: nearly all 40 samples fall into one giant cluster, with only a single sample splitting off on its own. This is the classic “chaining” behavior associated with single linkage, which tends to peel off one point at a time rather than forming balanced, meaningful groups.

Do the results depend on the type of linkage used? Yes, substantially. Complete and average linkage give broadly similar (though still imperfect) partial separations, while single linkage fails to recover any meaningful two-group structure at all.

(c) Which genes differ the most between the two groups?

A two-sample t -test was run for each of the 1,000 genes, comparing its expression values in the diseased group against the healthy group. Genes with the largest absolute t -statistic (equivalently, the smallest p -value) show the strongest evidence of a genuine mean difference between the two groups.

Gene	t -statistic	p -value
502	10.317	1.43×10^{-12}
589	10.025	3.19×10^{-12}
600	9.623	9.81×10^{-12}
590	8.976	6.28×10^{-11}
565	8.629	1.74×10^{-10}
551	8.011	1.10×10^{-9}
593	7.984	1.19×10^{-9}
584	7.877	1.65×10^{-9}
538	7.752	2.42×10^{-9}
569	7.717	2.69×10^{-9}

Conclusion: Genes 502, 589, and 600 show the largest absolute t -statistics, with extremely small p -values even after accounting for the fact that 1,000 tests were run simultaneously (a concern formally addressed by multiple testing correction, covered in Chapter 13). This t -test-based ranking directly answers the collaborator’s question and provides a starting point for further biological investigation into which genes are most associated with the disease.

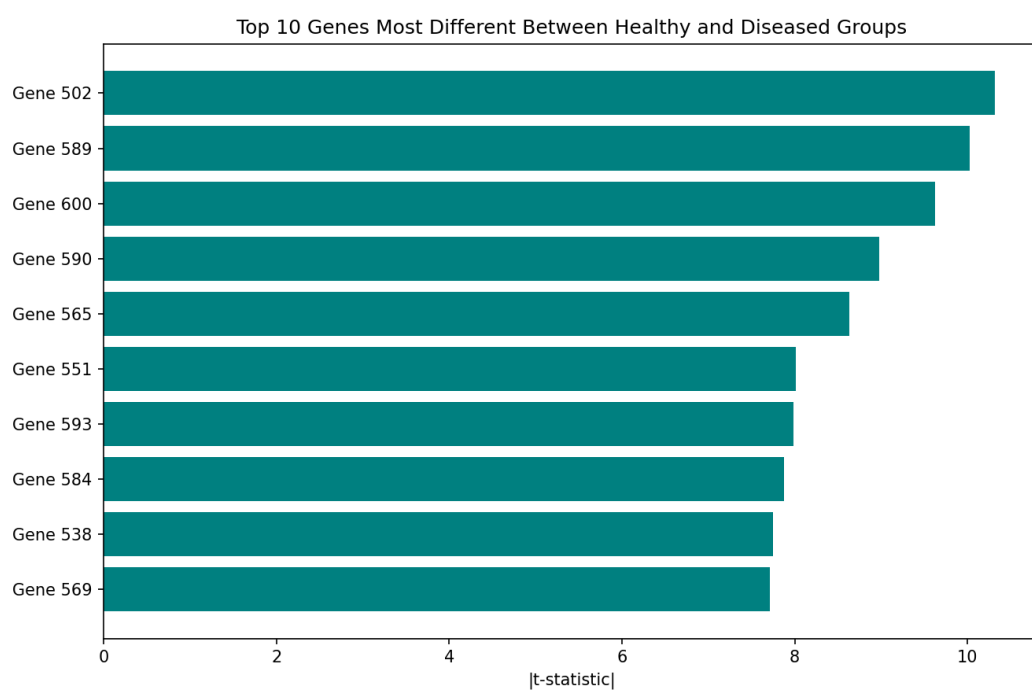


Figure 2: Top 10 genes ranked by absolute t-statistic between healthy and diseased groups.